HIGHER DIRECT IMAGES ON THE LISSE-ETALE (FLAT-FPPF) SITE AS LOCALIZATION COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Localization of a topos at a sheaf of sets. Sites, Lemma sites-lemma-localize-topos-site. Identifies the site and the localization morphism.
- Localization morphisms preserve injectives. Cohomology on Sites, Lemma sites-cohomology-lemma-cohomology-on-sheaf-sets. (pr')^{-1} preserves injectives.
- Pushforward on the localized site. Cohomology of Algebraic Stacks, Equation equation-pushforward-lisse-etale (in lemma-lisse-etale-functorial). Pointwise Gamma formula.

Source reference. Original lemma 07AT: sources/stacks-chow/tags.txt, source lines 9318–9318. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 07AU: sources/stacks-chow/tags.txt, source lines 9319–9319. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0791: sources/stacks-chow/tags.txt, source lines 9256–9256. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07A0: sources/stacks-chow/tags.txt, source lines 9290–9290. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *With assumptions and notation as in Lemma 07AT. Let $\mathcal{H}$ be an abelian sheaf on $\mathcal{X}_{lisse,étale}$ (resp. $\mathcal{X}_{flat,fppf}$). Then*

$$R^p f'_* \mathcal{H} = \text{sheaf associated to } y \longmapsto H^p((V \times_{y,\mathcal{Y}} \mathcal{X})', (pr')^{-1}\mathcal{H}) \tag{0.1.1}$$

Here y is an object of $\mathcal{Y}_{lisse,étale}$ (resp. $\mathcal{Y}_{flat,fppf}$) lying over the scheme V and the notation $(V \times_{y,\mathcal{Y}} \mathcal{X})'$ and pr' are explained in the proof.

Proof. As in the proof of Lemma 07AT let $(V \times_{y,\mathcal{Y}} \mathcal{X})' \subset V \times_{y,\mathcal{Y}} \mathcal{X}$ be the full subcategory consisting of objects (x, φ) where x is an object of $\mathcal{X}_{lisse,étale}$ (resp. $\mathcal{X}_{flat,fppf}$) and $\varphi : f(x) \to y$ is a morphism in $\mathcal{Y}$. By Equation (07AU) we have

$$f'_*\mathcal{H}(y) = \Gamma((V \times_{y,\mathcal{Y}} \mathcal{X})', \ (\mathrm{pr}')^{-1}\mathcal{H})$$

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where pr' is the projection. For an object (x, φ) of $(V \times_{y,\mathcal{Y}} \mathcal{X})'$ we can think of φ as a section of $(f')^{-1}h_y$ over x. Thus $(V \times_{\mathcal{Y}} \mathcal{X})'$ is the localization of the site $\mathcal{X}_{lisse,étale}$ (resp. $\mathcal{X}_{flat,fppf}$) at the sheaf of sets $(f')^{-1}h_y$, see Sites, Lemma 0791. The morphism

$$\mathrm{pr}' : (V \times_{y,\mathcal{Y}} \mathcal{X})' \to \mathcal{X}_{lisse,étale} \text{ (resp. } \mathrm{pr}' : (V \times_{y,\mathcal{Y}} \mathcal{X})' \to \mathcal{X}_{flat,fppf})$$

is the localization morphism. In particular, the pullback $(\mathrm{pr}')^{-1}$ preserves injective abelian sheaves, see Cohomology on Sites, Lemma 07A0.

Choose an injective resolution $\mathcal{H} \to \mathcal{I}^\bullet$ on $\mathcal{X}_{lisse,étale}$ (resp. $\mathcal{X}_{flat,fppf}$). By the formula for pushforward we see that $R^i f'_* \mathcal{H}$ is the sheaf associated to the presheaf which associates to y the cohomology of the complex

$$\begin{array}{c} \Gamma\Big((V \times_{y,\mathcal{Y}} \mathcal{X})', (\mathrm{pr}')^{-1}\mathcal{I}^{i-1}\Big) \\ \downarrow \\ \Gamma\Big((V \times_{y,\mathcal{Y}} \mathcal{X})', (\mathrm{pr}')^{-1}\mathcal{I}^{i}\Big) \\ \downarrow \\ \Gamma\Big((V \times_{y,\mathcal{Y}} \mathcal{X})', (\mathrm{pr}')^{-1}\mathcal{I}^{i+1}\Big) \end{array}$$

Since $(\mathrm{pr}')^{-1}$ is exact and preserves injectives the complex $(\mathrm{pr}')^{-1}\mathcal{I}^\bullet$ is an injective resolution of $(\mathrm{pr}')^{-1}\mathcal{H}$ and the proof is complete. $\square$